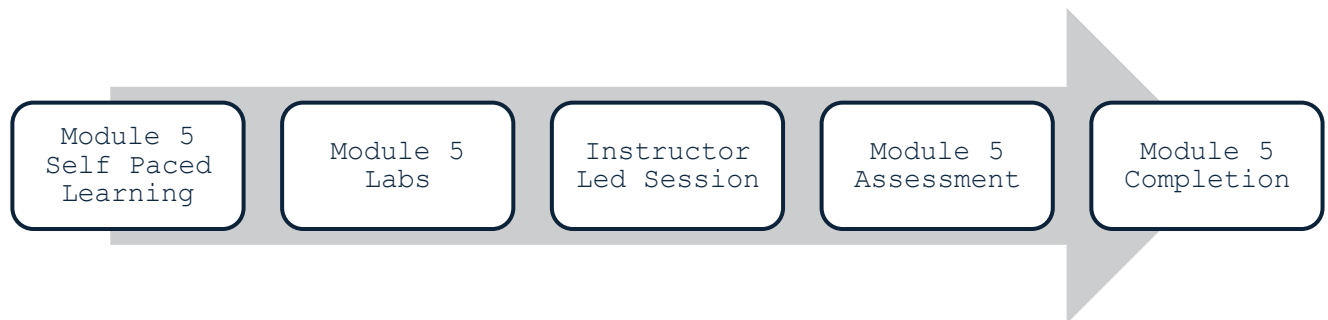


AI Pioneers: Claude Certified Architect – Foundations (CCA-F)

Module 5 — Context Management & Reliability

Overall Plan



Self-Paced Learning Plan

Module 5 introduces no new Skilljar courses — it builds on courses you completed in Modules 1–2. The supporting courses (and where you completed them) are listed below; revisit the specific lessons in the section-by-section plan. Sections 5 and 6 have no Skilljar course and are covered in the instructor-led lab.

Supporting course	Skilljar link
Introduction to Agent Skills — completed in Module 1	https://anthropic.skilljar.com/introduction-to-agent-skills
Introduction to Subagents — completed in Module 1	https://anthropic.skilljar.com/introduction-to-subagents
Claude Code in Action — completed in Module 2	https://anthropic.skilljar.com/claude-code-in-action

Key Sections: Context Management & Reliability

Sec	Section	Specific lesson(s) to watch
S1	Context Preservation & Optimization	• Introduction to Agent Skills: Configuration and multi-file skills (closest); optimization in lab
S2	Escalation & Ambiguity Resolution Patterns	• Introduction to Subagents: Using subagents effectively; Designing effective subagents • Introduction to Agent Skills: Skills vs. other Claude Code features (supporting)
S3	Error Propagation in Multi-Agent Systems	• Introduction to Subagents: Designing effective subagents; Using subagents effectively
S4	Context Management in Large Codebase Exploration	• Claude Code in Action: Adding context; Controlling context
S5	Human Review & Confidence Calibration	• No dedicated Skilljar lesson — built in the instructor-led lab
S6	Information Provenance & Uncertainty Handling	• No dedicated Skilljar lesson — built in the instructor-led lab

Labs

Lab 5.1 — Managing Context: Preservation, Optimization & Escalation (mandatory)

- Preserve persistent “case facts” so key details survive long sessions — e.g. keeping a customer's order ID pinned across turns.
- Trim and optimize tool outputs to protect the context window — e.g. summarizing a 500-row query into the 5 relevant fields.
- Escalate and resolve ambiguity rather than guessing — e.g. asking a clarifying question when two readings are plausible.

Lab 5.2 — Resilient Systems: Error Propagation & Large Codebase Exploration (mandatory)

- Propagate errors cleanly through multi-agent systems — e.g. a subagent failure surfacing to the coordinator instead of silently dropping.
- Explore large codebases with scratchpads to track findings — e.g. noting discovered file paths while mapping a monorepo.
- Use /compact and crash recovery to survive long sessions — e.g. compacting history before context overflow and resuming after a crash.

Lab 5.3 — Trust & Traceability: Human Review, Confidence & Provenance (Optional)

- Calibrate confidence and route low-confidence cases to humans — e.g. auto-approving high-confidence answers, escalating the rest.
- Link every claim to its source for traceability — e.g. attaching the exact document line behind each finding.
- Distinguish confirmed vs. contested findings — e.g. flagging a conclusion two sources disagree on for reviewer attention.

(Refer to the lab document over LMS for detailed information)

Instructor Led Session

Key Considerations before Instructor Led Session

Make sure you can answer these from your self-paced study:

- How do you keep critical details (IDs, dates) alive across a long session — a case-facts block, trimming tool outputs, and where you place critical info?
- When must you escalate (vs. proceed), and what goes into a structured handoff?
- How do you propagate errors in a multi-agent system — structured error types, retry-local, and distinguishing a timeout from a valid 'no data found'?
- What keeps a large-codebase Claude Code session from degrading — scratchpads, subagents, /compact, and state export?
How do confidence scores route work to human review, and how do you map each claim to its source (and handle conflicting sources)?

What the instructor-led session covers

Skilljar teaches the concepts; the 60-minute live lab is where you build and debug:

- Build a multi-turn workflow that persists case facts and trims tool outputs; compare quality before vs after (S1).
- Implement escalation logic with a structured handoff; resolve ambiguity instead of guessing (S2).
- Propagate structured errors from subagents to a coordinator with recovery logic (S3).
- Explore a large codebase using scratchpads, subagents, /compact, and crash recovery (S4).
- Add confidence scores and route low-confidence outputs to human review; do stratified sampling (S5).
- Map each claim to its source; keep confirmed vs contested findings with publication dates (S6).

Section-by-section study plan

S1 — Context Preservation & Optimization	
Key concepts	Long conversations lose details (numbers, dates). Trim tool outputs to relevant fields only. Keep a persistent 'case facts' block. Put critical info at the start or end of context.
Watch on Skilljar (specific lessons)	• Introduction to Agent Skills — Configuration and multi-file skills (supporting) Note: Closest lesson is the modular, persistent config block (≈ a case-facts block). Context-window optimization — trimming tool outputs and placing critical info at the start/end — is built in the lab.
Hands-on activity (try after watching)	Build a multi-turn workflow; extract & persist case facts; trim tool responses; compare quality before vs after optimization.
Suggested self-paced time	45 min

S2 — Escalation & Ambiguity Resolution Patterns	
Key concepts	Always escalate when the user asks — no exceptions. Also escalate on policy gaps or when stuck. Request clarification instead of guessing. Include a structured handoff with context.
Watch on Skilljar (specific lessons)	• Introduction to Subagents — Using subagents effectively; Designing effective subagents • Introduction to Agent Skills — Skills vs. other Claude Code features (supporting) Note: Escalation triggers and the structured handoff are built in the lab; the subagent lessons cover the coordination/handoff foundation.
Hands-on activity (try after watching)	Implement escalation logic; design a structured escalation response; simulate ambiguous cases; compare escalation strategies.
Suggested self-paced time	45 min

S3 — Error Propagation in Multi-Agent Systems

Key concepts	Propagate structured errors (type, partial results, alternatives). Distinguish timeout failures from a valid 'no data found'. Subagents retry locally; only send unresolved errors up.
Watch on Skilljar (specific lessons)	• Introduction to Subagents — Designing effective subagents; Using subagents effectively Note: <i>Structured error propagation (error type, partial results, retry-local, send only unresolved errors up) is built in the lab.</i>
Hands-on activity (try after watching)	Simulate subagent failures; implement structured error propagation; build coordinator recovery logic; add coverage-gap annotations.
Suggested self-paced time	45 min

S4 — Context Management in Large Codebase Exploration

Key concepts	Long Claude Code sessions degrade quality. Use scratchpad files for notes, subagents for isolated tasks, /compact to trim context, and state export for crash recovery.
Watch on Skilljar (specific lessons)	• Claude Code in Action — Adding context; Controlling context Note: <i>Scratchpad files, /compact, and state-export crash recovery are practised in the lab; these lessons cover the underlying context-control model.</i>
Hands-on activity (try after watching)	Explore a large codebase with Claude Code; create a scratchpad file; use subagents; apply /compact; simulate crash recovery.
Suggested self-paced time	60 min

S5 — Human Review & Confidence Calibration

Key concepts	Don't rely on aggregate accuracy — evaluate per field and per document type. Use confidence scores to route low-confidence outputs to human review. Use stratified sampling.
Watch on Skilljar (specific lessons)	• <i>No dedicated lesson in any CCA-F course — built in the instructor-led lab.</i> Note: No CCA-F course covers human review / confidence calibration. Closest optional resource: <i>AI Capabilities and Limitations</i>. See docs.claude.com for evaluation & calibration patterns. Built in the instructor-led lab.
Hands-on activity (try after watching)	Add confidence scores to outputs; design routing logic for human review; perform sampling-based evaluation; analyse accuracy segments.
Suggested self-paced time	30 min

S6 — Information Provenance & Uncertainty Handling

Key concepts	Map each claim to its source (URL, doc name, evidence excerpt). Keep conflicting info from multiple sources — don't pick one. Always include publication dates for temporal context.
Watch on Skilljar (specific lessons)	• <i>No dedicated lesson in any CCA-F course — built in the instructor-led lab.</i> Note: No CCA-F course covers provenance / uncertainty handling. Closest optional resource: <i>AI Capabilities and Limitations</i>. See docs.claude.com for provenance & uncertainty patterns. Built in the instructor-led lab.
Hands-on activity (try after watching)	Build a claim-source mapping output; process multi-source conflicting data; preserve attribution; generate a confirmed vs contested report.
Suggested self-paced time	30 min

